

Part III Seminar Abstracts - Lent 2026

Algebra

David He

Thursday, 2:20pm, MR5

Schur-Weyl duality and diagrammatics

Schur-Weyl duality is a family of results relating two commuting actions on tensor powers of a vector space. We will focus on the case of SL_2 and the symmetric group, which naturally leads to diagrammatics of the Temperley-Lieb category. Then we will discuss more general settings. This talk will be accessible to everyone with basic exposure to representation theory.

Algebraic Geometry

Brian O'Callaghan

Thursday, 3pm, MR5

Tropicalising hypergeometric tau-functions

Hypergeometric tau-functions are a priori purely analytic objects for computing solutions to certain differential equations. Surprisingly however, there are deep and rich links to combinatorics and further to the geometry of moduli spaces. We look at how the combinatorics and geometry interplay using tropical geometry to prove numerous structural properties of enumerative interest (for certain Hurwitz numbers)

Differential Geometry and Topology

Andrew Florescu

Thursday, 3:40pm, MR5

The regularity and singularity theory of area-minimising hypersurfaces

A very natural geometric question the problem of finding a surface of least area among all those spanning some fixed boundary. This came to be known as the 'Plateau' problem, following soap film experiments later conducted by a physicist of this name, and ultimately led to the development of minimal surface theory as a central theme in differential geometry. The first ever Fields medal was awarded to Douglas in 1936 for his work on the Plateau problem for two-dimensional surfaces in three dimensional space. It ultimately turns out however, that in ambient dimension at least 8, area minimising hypersurfaces with respect to smooth boundaries might not exist in the smooth class, and one must seek them in a rougher and more general class of submanifold-like objects arising from geometric measure theory. A great body of theory has developed since the late 20th century which has shown that the singular sets that arise in the minimisers for this problem must satisfy some very restrictive size and structural constraints - the goal of this talk is to survey some of these developments. Time permitting, I also hope to explore some very recent and exciting results that show that singularities in area minimising hypersurfaces can be generically perturbed away in dimensions up to 11, and how these ideas might have profound applications to a range of fields of study in modern geometric analysis, including mathematical general relativity, the topology of manifolds with positive scalar curvature, and geometric flows.

Number Theory

Elliot Ferguson

Thursday, 1pm, MR5

The Cohen-Lenstra heuristics

The class group of a number field is an important arithmetic invariant and a central object of study in number theory. The Cohen-Lenstra heuristics, formulated in the 1980s, are elegant predictions (which are still largely wide open) about the distribution of class groups over certain families of

number fields. We motivate and discuss the predictions for imaginary quadratic fields, which can be interpreted as asserting that class groups are essentially random objects, and briefly discuss some subsequent developments. Prior knowledge about class groups will not be necessary.

Naman Pratap

Thursday, 1.40pm, MR5

Iwasawa theory, Mazur--Tate elements and the Birch--Swinnerton-Dyer conjecture

We provide an overview of Iwasawa theory for elliptic curves, which has played a central role in many results towards the Birch and Swinnerton-Dyer (BSD) conjecture. Broadly, Iwasawa theory provides a 'p-adic' framework for studying arithmetic invariants (such as Selmer groups of elliptic curves) and relating them to values of analytic objects called L-functions.

We describe Mazur--Tate elements, which naturally arise in this framework and interpolate values of L-functions of elliptic curves and modular forms. We present our recent results on these elements which shed light on the BSD conjecture.

While some familiarity with elliptic curves will be useful, I will keep the prerequisites minimal.

Particle Physics, Quantum Fields and Strings

Nathan Godechal

Thursday, 1:40pm, MR15

Formal aspects of Chern-Simons theory

Discussion of Chern-Simons theory for compact $SU(2)$ gauge group (Wilson lines, quantization, functoriality, connections to Knots Theory). Followed by a discussion of Chern-Simons theory for non-compact $SL_2(\mathbb{C})$ gauge group (quantization, perturbative expansion, Habiro ring).

Vatsal

Thursday, 2:20pm, MR15

Green-Schwarz superstrings

Applied and Computational analysis

Joe Keeble

Thursday, 11:40pm, MR15

Neural networks for solving the Scalar Laplace equation on Calabi-Yau manifolds

Calabi-Yau Manifolds play an essential role in the dimensional compactification required for Superstring theory. However, no analytic solutions are known for the metrics of these spaces, which are essential if we are to determine their coupling to effective low-energy theories of physics and make phenomenological predictions. Numerical algorithms to determine the metric such as Donaldson's are effective but the computational cost scales factorially with increasing accuracy. Machine learning, and in particular Neural Networks, present a new, accurate, and significantly faster way to compute these metrics. These findings will be reviewed, and applied to modify an existing numerical algorithm, solving the Scalar Laplace equation with significant speed-up and comparable accuracy.

Combinatorics

Alex Karbowski

Thursday, 1pm, MR4

Markov Numbers and Christoffel Words

Let $p < q$ be coprime integers, and draw a line segment L from the origin $(0, 0)$ to the point (q, p) . Track the lattice points that lie immediately beneath L ; since L has slope $p/q < 1$, neighboring

lattice points either have the same y-coordinate (call this pattern A), or has y-coordinate differing by 1 (call this pattern B). Now consider the string of As and Bs you get going from left to right: for example, for $(q, p) = (3, 2)$ you get ABB. Strings generated this way are called Christoffel words, and this construction shows that Christoffel words can be identified with rational numbers between 0 and 1, via the slope of their corresponding segment. What properties do these Christoffel words have? We will answer this question by exhibiting a nice identification between Christoffel words and Markov numbers, integer solutions to the Diophantine equation $X^2 + Y^2 + Z^2 = 3XYZ$.

Nandi Wang

Thursday, 1:40pm MR4

Induced Poset Saturation

Given a finite poset P , how small can a family of subsets of $[n]$ be if it doesn't contain an induced copy of P , but adding any other subset creates one? Such families are called P -saturated, and the minimum size is known as the induced saturation number of P . We discuss techniques for bounding this number through several examples, then focus on the poset N and describe some interesting structural properties of N -saturated families that yield a linear lower bound on its induced saturation number.

Probability & Statistics

Jonathan Kelly

Thursday, 3pm MR4

Principled Approaches to Score Matching in Diffusion Models

Diffusion models are a central tool in generative AI, with particularly strong applications in image generation. They rely on a forward stochastic process that perturbs data and a reverse process that reconstructs it. The main challenge is to approximate the score function governing the reverse dynamics accurately enough to produce high-quality samples. In this seminar, I will present an alternative approach based on information propagation, where the density-inducing process is approximated from samples of the forward process rather than learned by a standard neural network model. We will also explore its qualitative and quantitative performance on some example datasets.

Zhichang (Dora) Chen

Thursday, 2.20pm MR4

Proximal causal inference: an (intuitive) overview

A famous phrase in statistics says: 'correlation does not imply causation'. Most traditional statistical theory focuses on prediction tasks, mostly relying on analysing the observed data distribution, but a fast-growing branch known as causal inference tries to answer causal questions, which are inherently harder. In this talk, I will briefly sketch the background of causal inference and give an intuitive overview of proximal causal inference, a recent approach to inferring causal effects when there is 'unmeasured confounding' present. This talk should be accessible to anyone with a basic background in statistics.

Analysis and PDEs

James Corcoran

Thursday, 4.20pm, MR5

From Hamiltonian Systems to Invariant Gibbs Measures

Gibbs measures are the central probability distributions in statistical and many-body physics and are also of great mathematical importance in the study of differential equations. This talk will recap the definition of the Gibbs measure on abstract measure spaces and will study its invariance properties in a certain class of ODE systems. Additionally, we will introduce the basics of Hamiltonian ODEs and how symmetry properties of solutions to these systems lead naturally to

invariance properties of the corresponding Gibbs measure, with the simple harmonic oscillator being the central example throughout.

Relativity and Cosmology

Elias Fink

Thursday, 10.20am, MR20

Black Hole Instabilities and Weak Cosmic Censorship

In four dimensions, uniqueness theorems strongly constrain the space of possible asymptotically flat solutions to the Einstein equation to the two dimensional Kerr family of black holes with horizon topology S^2 . In higher dimensions this is not the case and gravity becomes richer. We will focus on black strings and Myers-Perry black holes, a generalisation of Kerr to higher dimensions, to explore how perturbations of these black holes evolve in time and potentially lead to instabilities which can have detrimental effects on weak and strong cosmic censorship. Along the way, we will also discuss the mathematical techniques that are being employed to attack the hard problem of dynamical general relativity.

Luis Bonanad Estarlich

Thursday, 9:40am, MR20

The Rod Structure of General Relativity

By performing a Wick rotation to black hole spacetimes, one can find Riemannian manifolds solving Einstein's equations. Surprisingly, one finds thermodynamic quantities are encoded within the regularity of our solutions! This motivates the use of the gravitational path integral in the study of black holes, making the problem of characterising solutions to Einstein's equations relevant beyond pure mathematics. In this talk, I'll explore how rod structures arise from the canonical form of the metric and demonstrate their utility in constructing and interpreting exact solutions, with a particular focus on 4d gravitational instantons admitting a $U(1) \times U(1)$ isometry group.

Quantum Mechanics

Andreu Vega

Thursday, 3pm, MR15

Anomalous symmetry breaking in quantum mechanics

What happens when classical symmetries fail in the quantum world? This talk explores the transition from scale invariance to nonrenormalizability through attractive $1/r^N$ potentials.

We first examine the inverse square potential ($N=2$), where the breaking of classical scale symmetry via renormalization provides the simplest example of a quantum anomaly—a phenomenon evidenced by the critical dipole moment in polar molecules.

We then contrast this with the more singular $N > 2$ cases, arguing why these potentials require infinitely many parameters to be well-defined. By drawing a direct analogy with nonrenormalizable theories in QFT, we discuss the fundamental limits of predictability in singular quantum systems.

Condensed Matter Theory

Oisín Davey

Thursday, 3.40pm, MR15

Hofstadter's Butterfly

Overview of the Cantor set spectrum emerging from 2D magnetic materials. Topics: Peierls' substitution, Cocycles, Chern numbers, Interpretation, Integer Quantum Hall Effect.